

# **CYBER SECURITY TASK 01**

## **PART D – CYBER THREAT RESEARCH**

### **NAME : BHAKTI MAHADEV PARHAD**

#### **Introduction :**

Cyber threats are constantly evolving and pose significant risks to individuals, organizations, and governments. Understanding common cyber threats is essential for recognizing potential attacks and implementing appropriate security measures. This section explains five common cyber threats, provides real-world examples, and outlines effective prevention methods.

#### **1. Phishing**

**Definition :** Phishing is a cyberattack in which attackers send fraudulent emails, messages, or websites that appear legitimate to trick users into revealing sensitive information such as passwords, banking details, or personal data.

**Real-World Example :** An employee receives an email that appears to be from their bank asking them to verify their account by clicking a link. The link leads to a fake website that steals the user's login credentials.

#### **Prevention Methods :**

- Verify the sender before responding to emails.
- Avoid clicking suspicious links or downloading unknown attachments.
- Check website URLs carefully before entering sensitive information.
- Enable Multi-Factor Authentication (MFA).
- Report suspicious emails to the appropriate authority.

#### **2. Malware**

**Definition :** Malware is malicious software designed to damage, disrupt, or gain unauthorized access to computer systems. Common types include viruses, worms, spyware, and trojans.

**Real-World Example :** A user downloads a free software application from an untrusted website. The software secretly installs malware that steals personal information and slows down the computer.

**Prevention Methods :**

- Install trusted antivirus software.
- Keep the operating system and applications updated.
- Download software only from trusted sources.
- Avoid opening unknown email attachments.
- Perform regular system scans.

### **3. Ransomware**

**Definition :** Ransomware is a type of malware that encrypts files on a computer or network and demands payment to restore access.

**Real-World Example :** A company loses access to all of its important business files after a ransomware attack. The attackers demand payment in cryptocurrency to provide the decryption key.

**Prevention Methods :**

- Maintain regular backups of important files.
- Keep software and operating systems updated.
- Do not open suspicious emails or attachments.
- Use antivirus and endpoint protection software.
- Enable Multi-Factor Authentication where possible.

### **4. Social Engineering**

**Definition :** Social engineering is a technique used by attackers to manipulate people into revealing confidential information or performing actions that compromise security.

**Real-World Example :** An attacker pretends to be an IT support employee and convinces a staff member to share their login credentials to "fix" a technical problem.

**Prevention Methods :**

- Verify the identity of anyone requesting sensitive information.
- Never share passwords or OTPs.
- Participate in cybersecurity awareness training.
- Be cautious of urgent or unexpected requests.
- Follow organizational security policies.

**5. Data Breach**

**Definition :** A data breach occurs when confidential, personal, or sensitive information is accessed, disclosed, or stolen without authorization.

**Real-World Example :** A company's customer database is compromised, exposing users' names, email addresses, phone numbers, and passwords.

**Prevention Methods :**

- Use strong and unique passwords.
- Enable Multi-Factor Authentication (MFA).
- Encrypt sensitive data.
- Regularly update systems and software.
- Monitor accounts for unusual activity.

**Conclusion :**

Understanding common cyber threats helps individuals recognize potential risks and adopt safer online practices. By following cybersecurity best practices such as using strong passwords, enabling Multi-Factor Authentication, keeping systems updated, and remaining alert to suspicious activities, users can significantly reduce the risk of becoming victims of cyberattacks.